RECOMMENDATIONS

TO REDUCE THE COSTS FOR SIGMA TOYS

5 Distribution Route Optimization

Optimization of the route using Data Analysis Tool - SOLVER

Toy Manufacturing Location	Delhi	Chennai	Kolkata	Goa	Mumbai	Jaipur	Bhopal	Shimla
Noida	3713741	0	0	0	1766601	675248	233908	0
Nagpur	0	0	1519343	0	1418425	0	0	0
Bhubaneshwar	0	0	0	0	0	0	441388	0
Lucknow	0	0	0	0	898866	0	0	675200
Rajkot	0	1012848	0	506447	642647	0	0	0

- Route optimization is absolutely essential to **reduce cost for distribution**
- Sigma Toys should use freight cost, packaging cost and lead time as parameters to come up with an **optimum route**
- The total cost for packaging and distribution including opportunity and tax cost **after optimization** is 9.18 crores which is less than previous cost of 11 crores by **1.82 crores**

6 GEP SMART™

Source: GEP Software



- The solution will help Sigma Toys manage their increased supply of **raw material efficiently**
- The seamless integration from **sourcing to contracting** to order would reduce burden on employees and **improve working conditions** for them
- This will help the company get real time visibility on **cost saving initiatives** and help improve bottom line
- **Gig workers** can be monitored through contract management such that peak demand season is met with **adequate supply**

7 Reducing Employee Attrition

Sigma Toys can make the stand-by shift manager permanent and divide the workload the **among 4 managers**. This provided leave of **at least 2 days/ week** to each manager

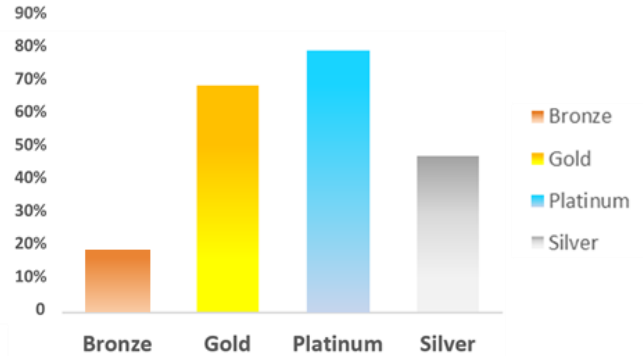
Gig Workers – Sigma Toys should introduce a **formal contract with Gig workers** during the **peak demand season** (March, April, October & November)

- As per the Industry standards, Sigma Toys pays **66% higher salary** to its employees. Thus, wage requirements are not the reason for **high attrition rate**
- As per Secondary research, we noted that the **working environment** plays an important for employee satisfaction .Employees seem to be working in a **24/7 shift** leading to high attrition

8 New Scrappage Policy

- Segmenting toys into 4 categories – **Bronze, Silver, Gold & Silver**, the toys can be **sold at varied prices** according to the segments they belong to
- Toys which do not meet the quality standards but still of **good quality** can be categorized as **Platinum** while Toys of **very poor quality** can be categorized as **Bronze**

Platinum Scrappage can be sold online and to dealers **without** Sigma Toys Label at **80% of the MRP**, while **Bronze** can be sold at **INR 50**



KEY ADDRESSABLE ISSUES

PROBLEMS WITH REVENUE

Sigma Toys lost an opportunity to earn an additional revenue of INR 108 Cr

The company was not able to meet the demand for 2020 by 21 lakh units

Low Capacity Utilization & Idle time for Equipment is not utilized well

- Throughput rate of **Reactor (R1)** is the lowest
- Thus, **R1** proves to be a **bottleneck** in the manufacturing process
- PVC production capacity per day is **lower than expected**

Throughput Rate for each Equipment				
Equipment	Output (MT PVC)	Processing time in hours	Waiting time in hours	Througput rate (MT PVC per hour)
R1 (Bottleneck)	355	4	4	44
C1	248	1	0	248
E1	213	0.5	0	426
D1	195	1.5	0	130
M1	177	1	0	177



- Scrapped Toys are sold at average INR 50, despite cost of toys amounting to approximately **INR 350**
- Company only uses **Retail Channel** for sales, **highly limiting** their revenue increasing options

PROBLEMS WITH COST



Bhubaneshwar has highest packaging cost, highest error rate & lowest efficiency, still it receives highest yearly PVC shipment increasing final cost

- **Packaging cost, error rate & taxes** of every toy manufacturing plant vary
- Despite this, all the plants run at **similar capacity utilization** of 60%. This increase the cost of the final product
- Transportation cost rising as **no Optimum route** from Toy manufacturing plant to distribution center has been found



- **Employee attrition is high**, due to hectic working schedule resulting in increased costs of hiring and training new joiners
- The company is facing **shelf availability issues** leading to opportunity cost
- Sigma Toys has **high error rate** at some plant locations

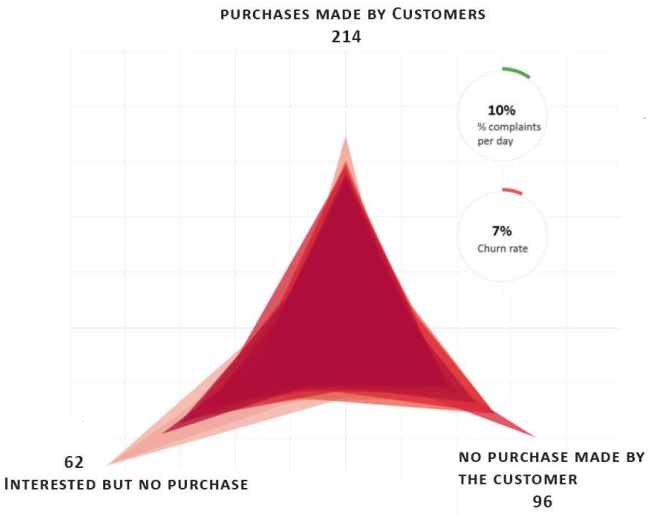
Gantt Chart highlighting inefficiencies in the PVC Manufacturing Plant. Click the chart to view the complete Day's Operations.

Legend: In Process, Idle, Cannot be Used

Equipment/ Time	12-12.30	12.30-1	1-1.30	1.30-2	2-2.30	2.30-3	3-3.30	3.30-4	4-4.30	4.30-5	5-5.30	5.30-6	6-6.30	6.30-7	7-7.30	7.30-8
R1																
C1																
E1																
D1																
M1																
Cycles	Cycle 1															

INSIGHTS GENERATED

IMPORTANT VARIABLES HIGHLIGHTING STRIKING INSIGHTS FROM ANALYSIS CONDUCTED



- **Majority of the students** evinced a **positive attitude** towards learning form **online courses**

- Students **preferred well-structured content** with recorded videos **uploaded in websites**

Quizzes and tests at the end of each course optimizes learning experience

- **Internet Connectivity** issues cited as **major setback** in learning

Personalization	User Adaptability	User Engagement	Product Guidance
45% Users	**37%** Users	**3X** Likely	**5X** Convenient

Source: Bain & Company

1 **Major Findings**

8/10 ***Users*** *Intrigued by the* ***Online Examinations***

- **Majority of students** preferred **quiz (75.9%) & assignments (56.3%)** at the end of **every class**

58% of learners want online classes twice a week with average duration of 45 min

- **Recorded classes** uploaded at **the university website/YouTube** was the most preferred

- Majority of respondents **(84%) preferred video content** supplemented with **reading materials**

DEVICES USED

Various **devices preferred to attend classes** - Smartphone **(57.98%)**, laptop **(35.83%)**, tablet **(4.89%)** and desktop **(0.65%)**

SUGGESTIONS

- **Convenience** cited as **biggest benefit**

Accessible from Learner's home, preferred space

- **Apt content,** easy follow-ups help **enhance learning**

To enhance the productivity of the learners, **long duration lectures should be avoided** and **sufficient breaks** should be **given between two consecutive lectures**

BUSINESS INSIGHTS

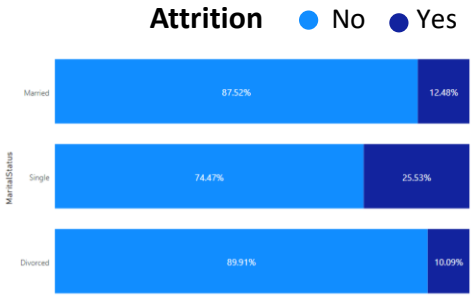
LOOKING AT IMPACT OF INCOME, JOB & ENVIRONMENT SATISFACTION OF THE EMPLOYEES

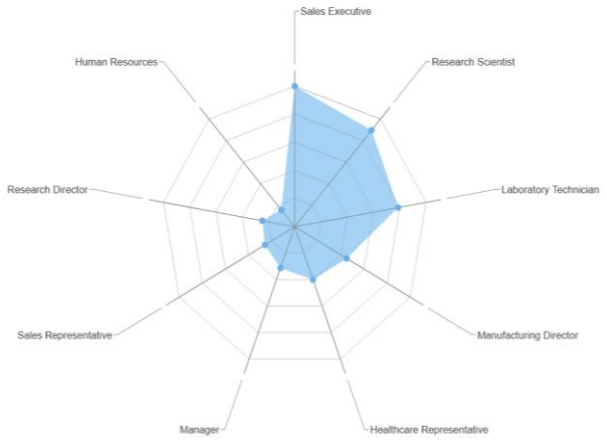
3 DECREASING ATTRITION OF SINGLES

Singles have high attrition rate due to multitude of reasons

2x higher Attrition Observed

7.14% more Hours by Single



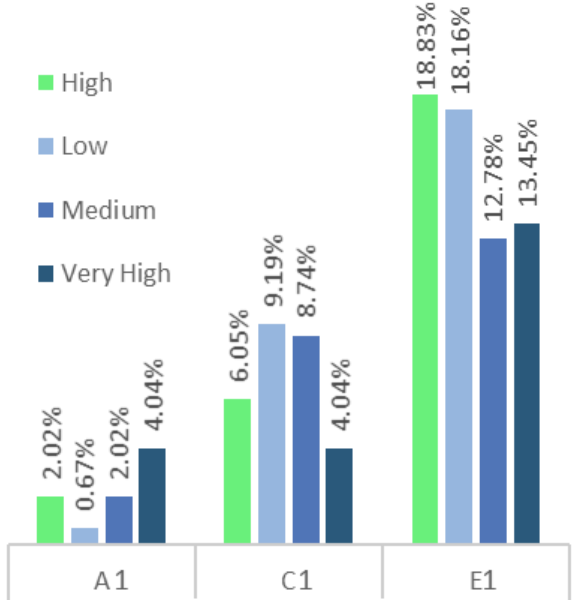


- **Singles have high attrition** rate with high & low level of job satisfaction with income > 50,000 for high, 12.27% is the attrition
- **Reason for High:** Switch for better job opportunities or already working in MNC

Apart from % hike, company needs to improve work culture, relations & environment, and rewards

- **Proper appraisals** & incentives, **Progressive HR policies**, **Positive working culture** and clarity over assigned role

4 IMPACT OF JOB & ENVIRONMENT SATISFACTION



Correlation = 0.77 for performance rating

- Class E1 people with **'higher job satisfaction'** & **'better work life balance'** are leaving.
- **Reason**: Looking for higher salary, Rigid company policies, looking for challenging work

RECOMMENDATION

- Better **performance ratings**
- **Flexible** company policies
- **Quality** output oriented
- Facilitate **Social Interaction**

5 INCREASING EMPLOYEE LONGEVITY

- **Incentive for younger people** to retain them for longer years
- **Integrating employees** to processes involving various elements
- **Develop 'Equity Plan'** to hook the younger employees
- Cater to other **personal needs of employees** too i.e., holidays etc.

Job clarity & well defined roles should be laid out clearly

Job evaluation exercise should be conducted

ULTIMATE SOLUTION

ENABLING A STEP CHANGE IN USER EXPERIENCE AND PERCEPTION WITH HOLISTIC EXPERIENCE

EAT confetti™

The BIG Idea

Phase 1 DEVELOPING EXPERIENCE → **Phase 2** USER PERSONALIZATION → **Phase 3** UNIQUELY FOR YOU

Store utilizes the holistic environment to use advantages offered online

Addressing pain points, and developing new services for **improved user experience**

Develop capabilities to **customize services** according to the **customer's preferences**

Developing online shopping experience | Loyal customer journey & new services | Personalized services & capabilities

1 Pehle Istemaal Kro Fir Vishwaas

Assistant #1



Want to taste confetti before purchase? Try here!

Available

Assistant #2



Instills sense of trust & increases convenience

Available

- **In-House section** where people can test or taste products before **making a purchase**
- Assistant counters will **increase customer engagement** & **enhance the shopping experience**
- Enables **enhanced personalization** for the **Eat Confetti consumer**

2 3P Model

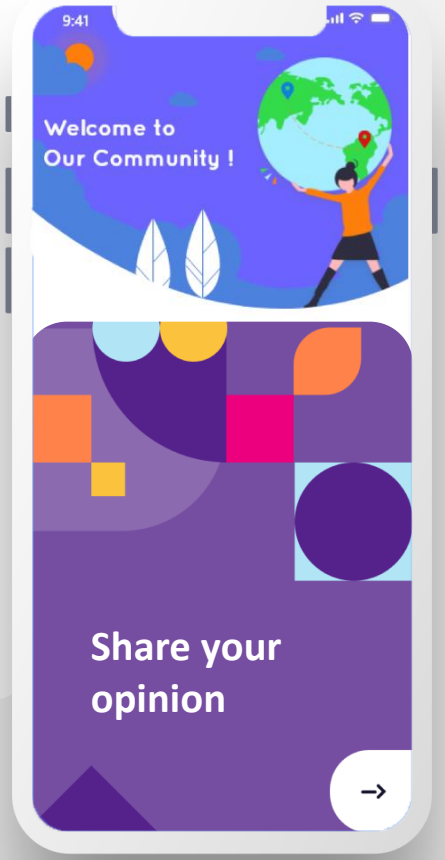


- Customers to receive tickets post every purchase
- Tickets collected and redeemed to win exciting gifts
- Engages customers for increased time within store

3 Benefits to Eat Confetti

IMPLEMENTING IDEA

Works as a loyalty program	Increases the consumer engagement	Better retention at the store	Holistic experience for customers



ULTIMATE SOLUTION

ENABLING A STEP CHANGE IN USER EXPERIENCE AND PERCEPTION TOWARDS PACKAGING

The BIG Idea

Phase 1 ENVIRONMENT IMPACT → **Developing Reusable Packaging** made of recycled material, upcycled at life end

Phase 2 CUSTOMER EXPERIENCE → **Multiple Size offers**, which are adjustable and comes with sealing solutions

Phase 3 SMART INVESTMENT → **Certified for ethics & sustainability** and comes at affordable price range

Usage of Reusable Packaging by users | **Improving Customer Experience** | **Better investment for all customers**

1 Benefits for Retailers



- Retailers can **replace single-use packaging** with our **sustainable EcoPack options**
- Once order is **packed & delivered**, customer can return prepaid **empty pack through mail**
- Packs are **checked, cleaned** and put **back to use**

2 Impact Created

Graphic Packaging INTERNATIONAL, amcor, Bemis, REXAM, Ball, DS Smith

- **Packs will be durable and reusable, reducing trash**
- **Amount of waste and CO2 emissions gets reduced**
- **Removes pollution waste management process need**

PREFERENCE OVER RECYCLING

Amount of plastic waste recycled	Accumulated In landfills, dumps, etc.	Tons of trash from recycling model	12 Billion Metric tons of plastic by	Plastic that has been incinerated
Only 9%	79% Waste	>5 Billion	Year 2050	Over 12%



INDUSTRY SCENARIO

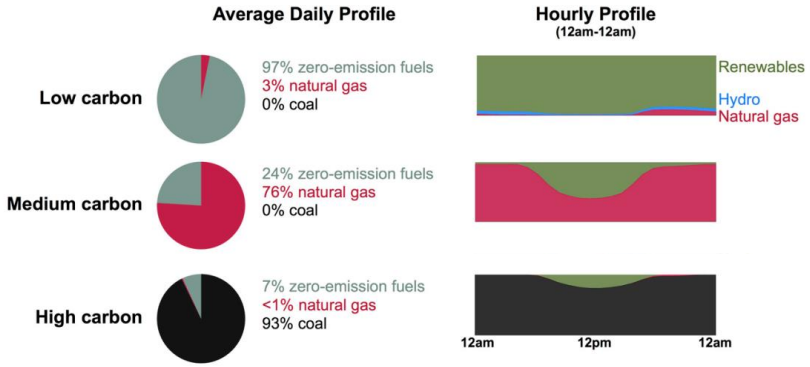
MARKET SCENARIO | IDENTIFICATION OF ISSUES | RECOMMENDATIONS | IMPLEMENTATION ROAD-MAP

KEY FACTS PERTAINING TO ELECTRONIC VEHICLES

Sources: WARC India, IBEF, Bain

200Km	15%	40%	B-100	2.5X
Can be travelled by Electric Vehicles	Market share of e-trucks by 2030	Fall in cost of EV batteries by 2027	Sale is the only mixture allowed	Increase in energy density by 2030

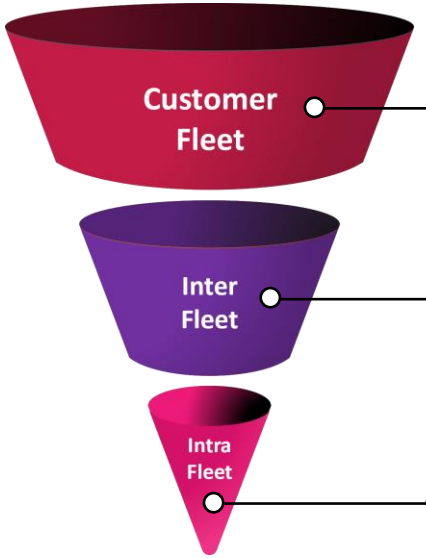
1 Innovative Technology



Levels of Carbon emission intensity based on fuel composition

- **B-100 biodiesel** is an EPA registered fuel shown to reduce emissions of **hydrocarbons, carbon monoxide & particulate**
- **Hybrid electric vehicles** are powered by internal combustion engine & electric motors, using **energy stored in batteries**
- **Ethanol blending with Flex Fuel Engines** is also a suggested **technology that can be adopted**

2 Alternative Fuels Adoption



Options Cited

- **Electric vehicles + Hybrid** for at least **300 Kms**
- **Hybrid (plug-in hybrid electric vehicles)** with **ICE of diesel** and **electric motors**
- **100% Electric vehicles** should be **used from the get go**

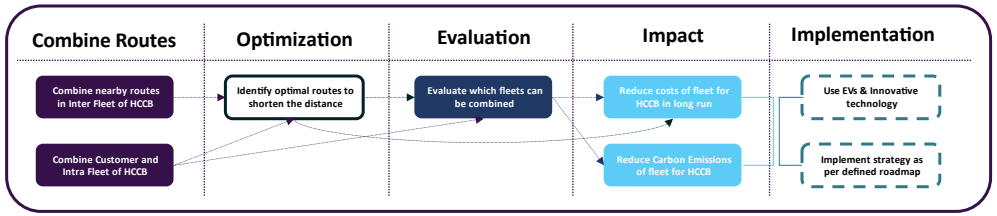
Strategy

Short term – Hybrid model as the **country-wide infrastructure** doesn't support EV for long distances

Long term – Shifting to **Biodiesel & Ethanol Blending**

EVs are the **cleanest source** of **energy in the sector**

3 Mapping Reroute Approach



4 Financial Impact

Type	Total Emission
Current Fleet	26730
PHEVS**	1485
Full Hybrids	3514.5
Pure Evs	0

High reduction in Carbon Emission can be observed

- **Further assumption** is made that all emissions are **Tailpipe Emissions**
- **Distance** Travelled is assumed to **be same**
- **PHEV** refers to **Plug-in Hybrid Vehicles**

Link to detailed analysis

CURRENT SCENARIO

OVERVIEW OF MARKET OPPORTUNITY AND OVERALL DESCRIPTION OF CAFÉ CHAINS

EAT confetti™

MARKET SCENARIO → IDENTIFICATION OF ISSUES → BUSINESS MODEL → IMPLEMENTATION ROAD-MAP

OVERALL MARKET DESCRIPTION OF CAFÉ CHAINS MARKET

Sources: WARC, Deloitte, Bain

$73bn
Market size of the café chains by the year 2024

7.84%
Forecasted 7.8% CAGR for the café chains

12.4%
Market growth rate from year 2019 to 2024

44.3%
Population to prefer touch and feel products

$1.7tn
Projected P2P café chains value for year 2027

93.1%
Customers prefer visiting cafes over eating at home

1 Customer Pain Points

884 — 1,200



***Word cloud** based on Tweets mentioning Physical Stores*

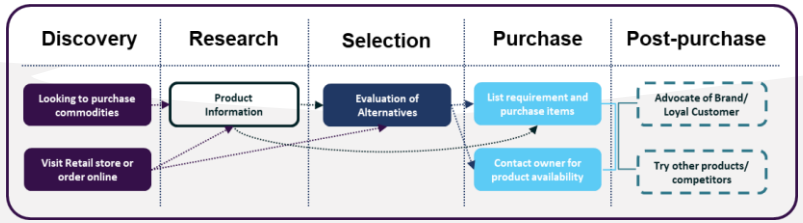
Long queues & limited choices, discourages customers from **visiting cafes & coffee shops**

- **Customers prefer** shopping from **comfort of homes**
- **Shopping hours** and **return/exchange policies** are **very rigid** when shopping online
- **Limited payment options** to choose from

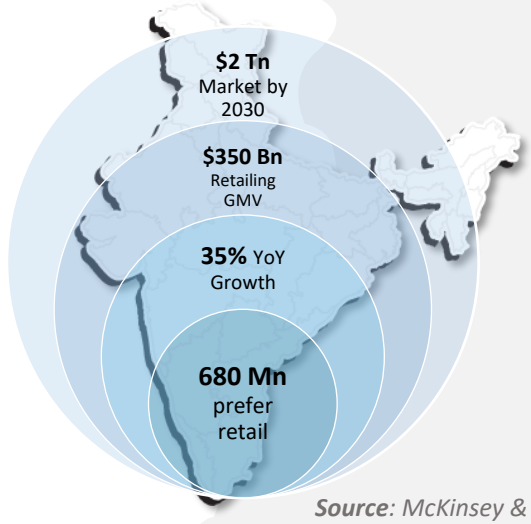
2 Identifying Customer

		Target Mkt	Strategy
New/ Non engaged customers	ToFu	Become part of **evoked set** in the **purchase decision**	Value through **affordable deals** to increase ticket size
Existing Low Loyalty customers	MoFu	Make Eat Confetti **first choice** for **purchasing,** over **competitors**	**Improve customer experience** for better engagement
Loyal & Engaged Customers	BoFu	**Simulate holistic experience** for **improving retention**	**Personalized communication** to increase word-of-mouth

3 Mapping Customer Journey



4 Estimating Market Size



Source: McKinsey & Company

- **Café chains to generate $2.5 bn by 2025**
- **India to be leader in café chains market**
- **Customer buying power to rise in future**